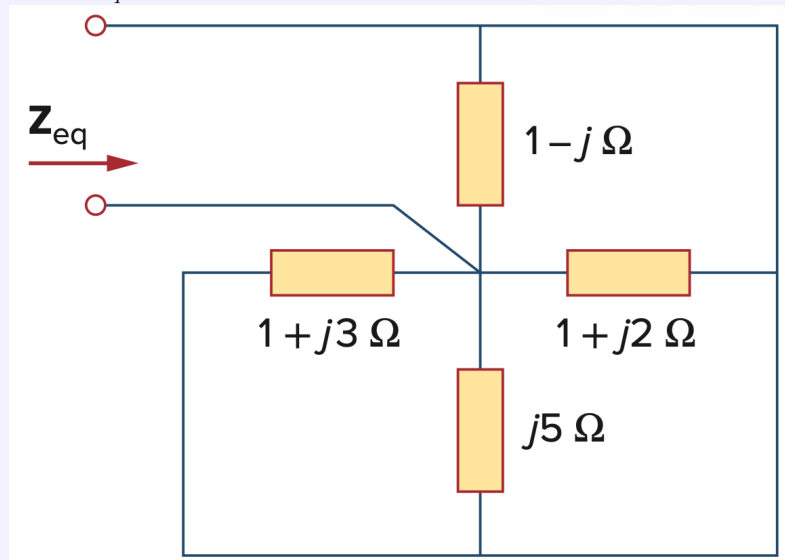


Exercise Sheet #7

AC 2

Exercise 1: Equivalent Impedance

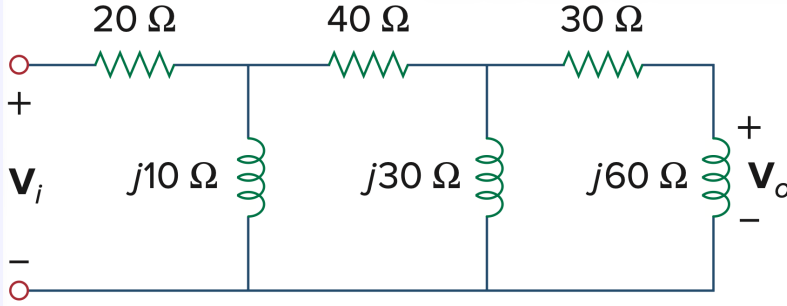
Find $\underline{Z}_{eq}$ in the circuit below.

Solution

$$\begin{aligned}\underline{Y} &= \frac{1}{\underline{Z}_1} + \frac{1}{\underline{Z}_2} + \frac{1}{\underline{Z}_3} + \frac{1}{\underline{Z}_4} = \frac{1}{1-j} + \frac{1}{1+j2} + \frac{1}{1+j3} + \frac{1}{j5} \\ &= \frac{1+j}{2} + \frac{1-j2}{5} + \frac{1-j3}{10} + \frac{-j5}{25} = \frac{25+10+5+j(25-20-15-10)}{50} = (0.8 - j0.4) \text{ S} \\ \underline{Z}_{eq} &= \frac{1}{\underline{Y}} = \frac{1}{0.8 - j0.4} = \frac{0.8 + j0.4}{0.8^2 + 0.4^2} = (1 + j0.5) \Omega\end{aligned}$$

Exercise 2: Stepwise Voltage Division

- Calculate the phase shift of the circuit below.
- State whether the phase shift is leading or lagging (output with respect to input).
- Determine the magnitude of the output when the input is 120 V.



Solution

- a) We take the input $\underline{v}_i$ as the reference phasor (angle 0°) and denote the node voltages by $\underline{v}_1, \underline{v}_2, \underline{v}_3$ (with $\underline{v}_3 = \underline{v}_o$).

Step 1: Relation between $\underline{v}_3$ and $\underline{v}_2$

Node 3 is a simple divider between the series resistor R_3 and the inductor $\underline{Z}_3$:

$$\underline{v}_3 = \underline{v}_2 \cdot \frac{\underline{Z}_3}{R_3 + \underline{Z}_3}.$$

With $R_3 = 30$, $\underline{Z}_3 = j60$:

$$\begin{aligned} \frac{\underline{v}_3}{\underline{v}_2} &= \frac{j60}{30 + j60} = \frac{1}{1 - j0.5} = 0.8 + 0.4j \\ &= \sqrt{0.8^2 + 0.4^2} \cdot \exp\left(j \arctan\left(\frac{0.4}{0.8}\right)\right) = 0.894427 \cdot \exp(j26.565^\circ). \end{aligned}$$

Step 2: Equivalent impedance seen from node 2

Node 2 sees $\underline{Z}_2$ in parallel with the series path $R_3 + \underline{Z}_3$:

$$\underline{Z}_{eq2} = \frac{\underline{Z}_2(R_3 + \underline{Z}_3)}{\underline{Z}_2 + (R_3 + \underline{Z}_3)}.$$

With $\underline{Z}_2 = j30$:

$$\begin{aligned} R_3 + \underline{Z}_3 &= 30 + j60, \\ \underline{Z}_{eq2} &= \frac{j30(30 + j60)}{j30 + (30 + j60)} = \frac{j900 - 1800}{30 + j90} = \frac{27000 + j189000}{9000} = 3 + j21. \end{aligned}$$

Now apply the divider between R_2 and $\underline{Z}_{eq2}$:

$$\begin{aligned} \underline{v}_2 &= \underline{v}_1 \cdot \frac{\underline{Z}_{eq2}}{R_2 + \underline{Z}_{eq2}}. \\ \frac{\underline{v}_2}{\underline{v}_1} &= \frac{3 + j21}{40 + 3 + j21} = \frac{(3 + j21)(43 - j21)}{43^2 + 21^2} = \frac{57 + j84}{229} \\ &= \frac{\sqrt{57^2 + 84^2}}{229} \cdot \exp\left(j \arctan\left(\frac{84}{57}\right)\right) = 0.443291 \cdot \exp(j55.840^\circ). \end{aligned}$$

Step 3: Equivalent impedance seen from node 1

Node 1 sees $\underline{Z}_1$ in parallel with the series path $R_2 + \underline{Z}_{eq2}$:

$$\underline{Z}_{eq1} = \frac{\underline{Z}_1(R_2 + \underline{Z}_{eq2})}{\underline{Z}_1 + (R_2 + \underline{Z}_{eq2})}$$

$$R_2 + \underline{Z}_{eq2} = 40 + 3 + j21 = 43 + j21,$$

$$\underline{Z}_1(R_2 + \underline{Z}_{eq2}) = j10(43 + j21) = -210 + j430$$

$$\underline{Z}_1 + (R_2 + \underline{Z}_{eq2}) = j10 + (43 + j21) = 43 + j31$$

Thus

$$\underline{Z}_{eq1} = \frac{-210 + j430}{43 + j31} = \frac{(-210 + j430)(43 - j31)}{43^2 + 31^2} = \frac{4300 + j25000}{2810}$$

$$= \sqrt{(4300^2 + 25000^2)} \cdot \exp(j \arctan(25000/4300)) \cdot \frac{1}{2810} = 9.027439 \cdot \exp(j80.241^\circ).$$

Then the divider between R_1 and $\underline{Z}_{eq1}$ gives:

$$\underline{v}_1 = \underline{v}_i \cdot \frac{\underline{Z}_{eq1}}{R_1 + \underline{Z}_{eq1}}.$$

$$R_1 + \underline{Z}_{eq1} = 20 + \frac{4300 + j25000}{2810} = \frac{20 \cdot 2810 + 4300 + j25000}{2810} = \frac{60500 + j25000}{2810}$$

$$= \sqrt{(60500^2 + 25000^2)} \cdot \exp(j \arctan(25000/60500)) \cdot \frac{1}{2810} = 23.296022 \cdot \exp(j22.452^\circ)$$

$$\frac{\underline{v}_1}{\underline{v}_i} = \frac{9.027439}{23.296022} \cdot \exp(j(80.241^\circ - 22.452^\circ)) = 0.387510 \cdot \exp(j57.789^\circ)$$

Step 4: Combine the three ratios

Output relative to input:

$$\begin{aligned} \frac{\underline{v}_o}{\underline{v}_i} &= \frac{\underline{v}_3}{\underline{v}_i} = \frac{\underline{v}_3}{\underline{v}_2} \cdot \frac{\underline{v}_2}{\underline{v}_1} \cdot \frac{\underline{v}_1}{\underline{v}_i} \\ &= 0.894427 \cdot \exp(j26.565^\circ) \cdot 0.443291 \cdot \exp(j55.840^\circ) \cdot 0.387510 \cdot \exp(j57.789^\circ) \\ &= 0.153644 \cdot \exp(j140.194^\circ) \\ &\Rightarrow \varphi = 140.2^\circ \end{aligned}$$

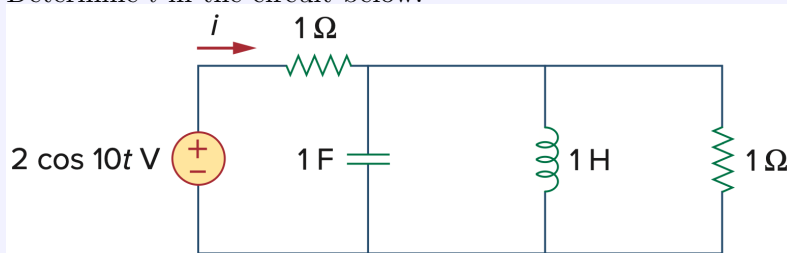
b) Since the phase shift is positive (+140.2°), the output phasor ****leads**** the input phasor.

c) Magnitude of the output when $V_i = 120 \text{ V}$:

$$V_o = 0.153644 \cdot 120 \text{ V} \approx 18.44 \text{ V}$$

Exercise 3: Nodal Analysis

Determine i in the circuit below.



Solution

$$v_s(t) = 2 \cos(10t) \Rightarrow \omega = 10$$

Capacitor and inductor admittances (1 F and 1 H):

$$\underline{Y}_C = j\omega C = j10, \quad \underline{Y}_L = \frac{1}{j\omega L} = -\frac{j}{10} = -j0.1,$$

and the parallel resistor gives $R = 1$. Thus the total admittance of the parallel block is

$$\underline{Y} = R + \underline{Y}_C + \underline{Y}_L = 1 + j10 - j0.1 = 1 + j9.9.$$

KCL at the node (current through series 1Ω equals current into the parallel block):

$$\frac{\hat{v}_s - \hat{v}}{1} = \hat{v} \underline{Y} \Rightarrow \hat{v} (1 + \underline{Y}) = \hat{v}_s \Rightarrow \hat{v} = \frac{\hat{v}_s}{1 + \underline{Y}}.$$

The source current $\hat{i}$ (which is i in time domain) is the current through the series 1Ω :

$$\hat{i} = \frac{\hat{v}_s - \hat{v}}{1} = \hat{v}_s \frac{\underline{Y}}{1 + \underline{Y}}$$

Substitute $\underline{Y} = 1 + j9.9$:

$$\hat{i} = 2 \cdot \frac{1 + j9.9}{2 + j9.9} = \frac{200.02 + j19.8}{102.01} \approx (1.9608 + j0.1941) \text{ A}$$

Magnitude, phase and back to the time domain (cosine reference):

$$i(t) = 1.9704 \cos(10t + 5.655^\circ) \text{ A}$$

Exercise 4: Instantaneous and Average Power

If $v(t) = 160 \cos(50t) \text{ V}$ and $i(t) = -33 \sin(50t - 30^\circ) \text{ A}$, calculate the instantaneous power and the average power.

Solution

First rewrite the current using

$$\sin x = \cos(x - 90^\circ) :$$

$$i(t) = -33 \cos[(50t - 30^\circ) - 90^\circ] = -33 \cos(50t - 120^\circ).$$

The instantaneous power is

$$p(t) = v(t)i(t) = 160 \cos(50t) [-33 \cos(50t - 120^\circ)] = -5280 \cos(50t) \cos(50t - 120^\circ).$$

Use the identity

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)].$$

Let $A = 50t$, $B = 50t - 120^\circ$:

$$p(t) = -5280 \cdot \frac{1}{2} [\cos(120^\circ) + \cos(100t - 120^\circ)] = -2640 [\cos 120^\circ + \cos(100t - 120^\circ)].$$

Since $\cos 120^\circ = -\frac{1}{2}$,

$$p(t) = -2640 \left[-\frac{1}{2} + \cos(100t - 120^\circ) \right] = 1320 - 2640 \cos(100t - 120^\circ).$$

Thus the **instantaneous power** is

$$p(t) = 1320 + 2640 \cos(100t + 60^\circ) \text{ W}.$$

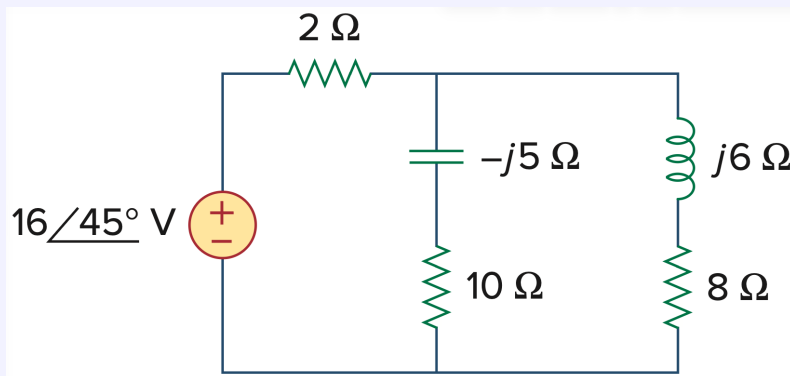
The **average power** is the DC term:

$$\bar{P} = 1320 \text{ W}.$$

Exercise 5: Complex Power Calculation

For the entire circuit below, calculate:

- the complex power
- the average power delivered by the source
- the reactive power
- the apparent power
- the power factor



Solution

Two parallel branches after the 2Ω resistor:

$$\underline{Z}_1 = 10 - j5, \quad \underline{Z}_2 = 8 + j6.$$

Total impedance (including the 2Ω resistor in series):

$$\begin{aligned} \underline{Z} &= 2 + \left(\frac{1}{\underline{Z}_1} + \frac{1}{\underline{Z}_2} \right)^{-1} = 2 + \left(\frac{1}{10 - j5} + \frac{1}{8 + j6} \right)^{-1} \\ &= 2 + \left(\frac{10 + j5}{100 + 25} + \frac{8 - j6}{64 + 36} \right)^{-1} = 2 + \left(0.08 + j0.04 + 0.08 - j0.06 \right)^{-1} \\ &= 2 + \frac{0.16 + j0.02}{0.16^2 + 0.02^2} = 2 + \frac{0.16 + j0.02}{0.026} = \frac{0.212 + j0.02}{0.026} \Omega \\ Z &= \frac{\sqrt{(0.212^2 + 0.02^2)}}{0.026} \approx 8.1901 \Omega \\ \varphi &= \arctan \frac{0.02}{0.212} = 5.39^\circ \end{aligned}$$

Source voltage (phasor complex RMS value):

$$\underline{V}_s = 16 \cdot e^{j45^\circ} = (11.3137 + j 11.3137) \text{ V}$$

Source current (phasor complex RMS value):

$$\underline{I}_s = \frac{\underline{V}_s}{\underline{Z}} = \frac{16}{8.1901} \cdot e^{j(45^\circ - 5.39^\circ)} = 1.9536 \cdot e^{j39.61^\circ} = (1.5051 + j 1.2455) \text{ A}$$

a) Complex power delivered by the source:

$$\underline{S} = \underline{V}_s \underline{I}_s^* = (11.3137 + j 11.3137) \text{ V} \cdot (1.5051 - j 1.2455) \text{ A} = (31.12 + j 2.94) \text{ VA}.$$

b) Active power delivered by source:

$$P = \text{Re}(\underline{S}) = 31.12 \text{ W}$$

c) Reactive power:

$$Q = \text{Im}(\underline{S}) = 2.94 \text{ VAR}$$

d) Apparent power:

$$S = \sqrt{P^2 + Q^2} = 31.26 \text{ VA}.$$

e) Power factor (scalar) and type:

$$\cos \varphi = \frac{P}{S} = \frac{31.12}{31.26} \approx 0.9956.$$

Since $Q > 0$ the net load is inductive, so the power factor is *lagging*.

Exercise 6: RMS Values and Average Power

The voltage across a load and the current through it are given by $v(t) = 20 + 60 \cos(100t)$ V and $i(t) = 1 - 0.5 \sin(100t)$ A. Find:

- the RMS values of the voltage and of the current
- the average power dissipated in the load

Solution

a) For a signal composed of a constant (DC) term plus a single sinusoid,

$$x(t) = X_{\text{DC}} + \hat{x} \cos(\omega t + \phi),$$

the mean square (time average of $x^2(t)$) over one period is

$$\overline{x^2(t)} = X_{\text{DC}}^2 + \frac{\hat{x}^2}{2},$$

because the cross term between DC and the sinusoid averages to zero and the sinusoid contributes $\hat{x}^2/2$. Thus the RMS value is

$$X = \sqrt{X_{\text{DC}}^2 + \frac{\hat{x}^2}{2}}.$$

Apply this to the voltage:

$$V_{\text{DC}} = 20, \quad \hat{v} = 60$$

Hence

$$V = \sqrt{20^2 + \frac{60^2}{2}} = \sqrt{2200} = 10\sqrt{22} \approx 46.9 \text{ V.}$$

Apply the same formula to the current. Write the current as

$$i(t) = I_{\text{DC}} + \hat{i} \cos(100t + \phi_i).$$

Here $I_{\text{DC}} = 1$. The AC term is $-0.5 \sin(100t)$, which has amplitude $\hat{i} = 0.5$. (The sign and phase do not affect the RMS magnitude.) Thus

$$I = \sqrt{1^2 + \frac{(0.5)^2}{2}} = \sqrt{1.125} \approx 1.061 \text{ A.}$$

b) The instantaneous power is

$$p(t) = v(t)i(t) = (20 + 60 \cos(100t))(1 - 0.5 \sin(100t)).$$

Expand:

$$\begin{aligned} p(t) &= 20 \cdot 1 + 20 \cdot (-0.5 \sin(100t)) + 60 \cos(100t) \cdot 1 + 60 \cos(100t) \cdot (-0.5 \sin(100t)) \\ &= 20 - 10 \sin(100t) + 60 \cos(100t) - 30 \cos(100t) \sin(100t) \end{aligned}$$

The average power P is the time average of $p(t)$ over one period. Every pure sinusoid (or product of sinusoids of the same frequency that reduces to a sinusoid at twice the frequency) has zero average. Thus the time average of $-10 \sin(100t)$, of $60 \cos(100t)$, and of $-30 \cos(100t) \sin(100t)$ is zero. Therefore only the DC term remains:

$$P = \overline{p(t)} = 20 \text{ W}$$

Alternatively, using the decomposition into DC and sinusoidal parts:

$$P = V_{\text{DC}} I_{\text{DC}} + \frac{1}{2} \hat{v} \hat{i} \cos \varphi,$$

where φ is the phase difference between the voltage and current sinusoids. Here $V_{\text{DC}} I_{\text{DC}} = 20 \cdot 1$ and the sinusoidal part has

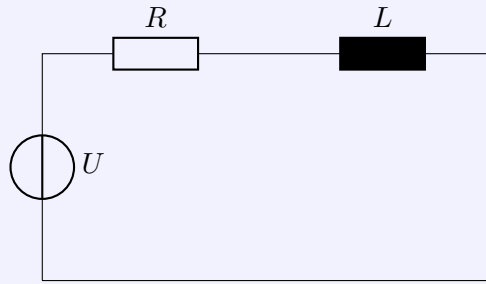
$$\hat{v} = 60, \quad \hat{i} = 0.5.$$

Voltage sinusoid: $60 \cos(100t)$ (phase 0). Current sinusoid: $-0.5 \sin(100t) = 0.5 \cos(100t + \frac{\pi}{2})$ (phase $+\frac{\pi}{2}$). Thus $\varphi = 0 - (+\frac{\pi}{2}) = -\frac{\pi}{2}$ and $\cos \varphi = \cos(-\frac{\pi}{2}) = 0$. Hence the sinusoidal contribution is zero and

$$P = 20 \text{ W.}$$

Exercise 7: Complex Network Analysis

Given the following network with a sinusoidal voltage source $U = 230 \text{ V}$, $f = 50 \text{ Hz}$ and the components $R = 20 \Omega$, $L = 100 \text{ mH}$:



Determine:

1. the complex impedance Z ,
2. the RMS current I ,
3. the phase shift between current and voltage,
4. the power factor $\cos \varphi$,
5. the active power P ,
6. the reactive power Q .

Solution

$$\omega = 2\pi f = 314 \text{ rad/s}$$

$$X_L = \omega L = 31.4 \Omega$$

$$\underline{Z} = (20 + j31.4) \Omega \Rightarrow |\underline{Z}| = 37.2 \Omega$$

$$I = \frac{230}{37.2} = 6.18 \text{ A}$$

$$\varphi = \arctan\left(\frac{31.4}{20}\right) = 57.5^\circ$$

$$\cos \varphi = 0.54$$

$$P = UI \cos \varphi = 764 \text{ W}$$

$$Q = UI \sin \varphi = 1199 \text{ VAR}$$

Exercise 8: Reactive Power Compensation

An AC motor with impedance $\underline{Z} = 4.2 + j 3.6 \Omega$ is supplied by a 220-V, 60-Hz source.

- a) Find the power factor, the real power P , and the reactive power Q .
- b) Determine the capacitor required to be connected in parallel with the motor so that the power factor is corrected to unity.

Solution

- a) The magnitude and angle of the load impedance are

$$|\underline{Z}| = \sqrt{R^2 + X^2} = \sqrt{4.2^2 + 3.6^2} \approx 5.532 \Omega,$$

$$\varphi = \arctan \frac{X}{R} = \arctan \frac{3.6}{4.2} \approx 0.709 \text{ rad} \approx 40.6^\circ.$$

The RMS current is

$$I = \frac{V}{|Z|} = \frac{220}{5.532} \approx 39.77 \text{ A}.$$

The power factor (PF) is the cosine of the impedance angle (lagging because $X > 0$):

$$\cos \varphi \approx 0.76 \quad (\text{lagging}).$$

Compute the complex power:

$$\underline{S} = \underline{V}\underline{I}^* = V \frac{V^*}{\underline{Z}^*} = \frac{V^2}{\underline{Z}^*} = V^2 / (R - jX)^* = \frac{V^2(R + jX)}{R^2 + X^2}$$

So the real and reactive powers are

$$P = \frac{V^2 R}{R^2 + X^2} \approx 6.643 \text{ kW},$$

$$Q = \frac{V^2 X}{R^2 + X^2} \approx 5.694 \text{ kVAR}.$$

- b) To correct the power factor to unity we must supply capacitive reactive power Q_C that cancels the motor's inductive reactive power Q . For a capacitor connected in parallel across the same RMS voltage V ,

$$|Q_C| = V^2 \omega C, \quad \omega = 2\pi f.$$

We need $|Q_C| = Q$, hence

$$C = \frac{Q}{V^2 \omega} = \frac{5694}{220^2 \cdot 2\pi \cdot 60} \approx 312 \mu\text{F}.$$

Exercise 9: Series Resonance

A series RLC network has $R = 2 \text{ k}\Omega$, $L = 40 \text{ mH}$, and $C = 1 \mu\text{F}$. Calculate the impedance at resonance and at one-fourth, one-half, twice, and four times the resonant frequency.

Solution

For a series RLC network with

$$R = 2 \text{ k}\Omega, \quad L = 40 \text{ mH}, \quad C = 1 \mu\text{F}$$

the series impedance is

$$\underline{Z}(j\omega) = R + j(\omega L - \frac{1}{\omega C}).$$

Resonance occurs when the net reactance is zero:

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute the values:

$$\omega_0 = \frac{1}{\sqrt{(40 \times 10^{-3})(1 \times 10^{-6})}} = 5000 \text{ rad s}^{-1}$$

For a given angular frequency ω ,

$$X_L = \omega L, \quad X_C = \frac{1}{\omega C}, \quad X = X_L - X_C,$$

$$\underline{Z} = R + jX, \quad |\underline{Z}| = \sqrt{R^2 + X^2}, \quad \varphi = \arctan \frac{X}{R}.$$

We evaluate at $\omega = k\omega_0$ for $k = \frac{1}{4}, \frac{1}{2}, 1, 2, 4$.

k	ω (rad/s)	X_L (Ω)	X_C (Ω)	$\underline{Z}$ (k Ω)	$ \underline{Z} $ (k Ω)	φ
0.25	1250	50	800	$2 - j0.75$	2.136	-20.6°
0.5	2500	100	400	$2 - j0.3$	2.022	-8.5°
1	5000	200	200	2	2	0°
2	10000	400	100	$2 + j0.3$	2.022	$+8.5^\circ$
4	20000	800	50	$2 + j0.75$	2.136	$+20.6^\circ$

Observations:

- At resonance ($k = 1$) the inductive and capacitive reactances cancel, leaving purely resistive impedance $Z = R = 2000 \Omega$.
- The network is capacitive ($X < 0$) for frequencies below resonance and inductive ($X > 0$) for frequencies above resonance.
- The magnitudes at pairs $k = \frac{1}{4}$ and $k = 4$ are equal (symmetry), and likewise for $k = \frac{1}{2}$ and $k = 2$.

Exercise 10: Parallel Resonance

Rework Exercise 6 if the elements are connected in parallel.

Solution

For the parallel connection the total admittance is

$$\underline{Y}(j\omega) = \underbrace{\frac{1}{R}}_G + j \left(\underbrace{\omega C}_{B_C} + \underbrace{\left(-\frac{1}{\omega L}\right)}_{B_L} \right) = G + jB(\omega).$$

The impedance is the reciprocal of the admittance:

$$\underline{Z}(\omega) = \frac{1}{\underline{Y}(\omega)} = \frac{1}{G + jB} = \frac{G - jB}{G^2 + B^2}$$

Its magnitude and phase are

$$|\underline{Z}| = \frac{1}{\sqrt{G^2 + B^2}}, \quad \varphi = \text{atan2}(\text{Im}(\underline{Z}), \text{Re}(\underline{Z})) = \arctan \frac{-B}{G}.$$

Parallel resonance (net susceptance zero) occurs at the same angular frequency as series resonance:

$$B(\omega_0) = 0 \quad \Rightarrow \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

With the given values

$$\omega_0 = \frac{1}{\sqrt{(40 \times 10^{-3})(1 \times 10^{-6})}} = 5000 \text{ rad s}^{-1}.$$

Also

$$G = \frac{1}{R} = \frac{1}{2000} = 5 \times 10^{-4} \text{ S}.$$

At resonance $B(\omega_0) = 0$, so

$$Y(\omega_0) = G \quad \Rightarrow \quad Z(\omega_0) = \frac{1}{G} = R = 2000 \Omega.$$

We compute for each ratio k :

$$\omega = k\omega_0, \quad B = \omega C - \frac{1}{\omega L}, \quad \underline{Z} = \frac{G - jB}{G^2 + B^2}$$

k	ω (rad/s)	B_C (m Ω)	B_L (m Ω)	$\underline{Z}$ (Ω)	$ \underline{Z} $ (Ω)	φ
0.25	1250	1.25	20	$1.42 + j 53.30$	53	$+88.5^\circ$
0.5	2500	2.5	10	$8.85 + j 132.74$	133	$+86.2^\circ$
1	5000	5	5	2000	2000	0°
2	10000	10	2.5	$8.85 - j 132.74$	133	-86.2°
4	20000	20	1.25	$1.42 - j 53.30$	53	-88.5°

Notes on the results:

- At resonance the susceptances of the inductor and capacitor cancel and the parallel branch behaves as a pure conductance G , giving $Z(\omega_0) = R$.
- For frequencies below resonance ($k < 1$) the net susceptance B is negative (inductive behavior), producing an impedance with a large positive imaginary part (inductive reactance).
- For frequencies above resonance ($k > 1$) B is positive (capacitive behavior), producing impedance with a large negative imaginary part (capacitive reactance).
- The impedances occur in conjugate pairs for reciprocal frequency ratios (e.g. $k = 0.25$ and $k = 4$; $k = 0.5$ and $k = 2$).